AI 500 - Foundations of AI

QUIZ 5

November 8, 2025

Name: _____ **Roll Number:** _____

- Each question may have **multiple correct answers**. Select all options that apply.

Q1. Which of the following statements about **association and causation** are true?

A. Every causal relationship implies an association.

B. Every association implies a causal relationship.

C. A strong correlation between X and Y always means X causes Y.

D. Two variables can be associated because of a hidden confounder.

E. Causation can exist even if correlation is weak or zero.

Q2. Which of the following are true about the **potential outcomes framework**?

A. $Y_i(1)$ is observed for treated individuals and $Y_i(0)$ is their counterfactual.

B. $Y_i(0)$ is observed for controlled individuals and $Y_i(1)$ is their counterfactual.

C. The causal effect for individual i is $\tau_i = Y_i(1) - Y_i(0)$.

D. The fundamental problem of causal inference arises because we can observe both $Y_i(1)$ and $Y_i(0)$.

E. The Average Treatment Effect (ATE) is $E_i[Y_i(1)] - E_i[Y_i(0)]$.

Q3. Which of the following can help ensure that the **difference-in-means estimator equals the Average Treatment Effect (ATE)**?

A. Random assignment of treatment.

B. Ignorability — potential outcomes are independent of treatment.

C. Presence of confounding variables.

D. SUTVA (Stable Unit Treatment Value Assumption).

E. Heterogeneous treatment effects across all units.

Q4. Which of the following statements about **confounding** are true?

A. A confounder is a variable that influences both treatment and outcome.

B. Controlling for confounders can reduce bias in causal estimates.

C. Randomization eliminates confounding on average.

D. Confounding can make an association appear causal when it isn't.

E. Confounding always decreases the observed association strength.

Q5. Which of the following correctly identify **causal inference as a missing data problem**?

A. We can observe both potential outcomes, but only for some individuals.

B. The counterfactual outcome is unobserved for each unit.

C. Randomization allows us to estimate the missing counterfactual statistically.

D. Quasi-experimental methods attempt to reconstruct missing counterfactuals.

E. The missing outcomes are unimportant for causal analysis.

Q6. Which of the following statements is the **most correct interpretation** of the claim

$$P(Y \mid X) > P(Y)$$

in the context of causal inference?

A. It proves that X causes Y.

B. It shows that X and Y are statistically associated, but does not prove that X causes Y.

C. It implies that Y causes X.

D. It confirms that there is no confounding between X and Y.

E. It demonstrates that X and Y are independent.

Q7. Estimating ATE vs. Difference-in-Means (D)

You are evaluating whether a training program improves test scores. For each person, $Y_i(1)$ represents the test score if they took the training, and $Y_i(0)$ represents the score if they did not. Each person either received the training ($T_i = 1$) or did not ($T_i = 0$), but not both.

Person (i)	**Treatment** T_i	$Y_i(1)$	$Y_i(0)$
1	1	85	70
2	1	78	75
3	0	80	68
4	0	74	72
5	1	90	83
6	0	77	74

Tasks:

(a) Compute the **Average Treatment Effect (ATE)**.(1)

(b) Compute the **difference-in-means (D)** estimator.(1)

(c) Compare your results for ATE and D.(2)